

4. ①可交换: $\forall x, y \in I, x * y = x + y - xy$

$$y * x = y + x - yx = x + y - xy$$

②可结合: $\forall x, y, z \in I, x * y * z = (x + y - xy) * z$

$$= x + y - xy + z - (x + y - xy)z = x + y + z - xy - yz - xz + xyz$$

$$= x * (y * z)$$

③单位元. 即找到 e 使得 $\forall x \in I, x * e = e * x = x$.

$$\text{由可交换. 只需 } x * e = x + e - ex = x. \quad e(1-x) = 0$$

$$\therefore e = 0$$

④逆元, $\forall x$ 可逆, $\exists x^{-1} \in I, x * x^{-1} = e = 0$

$$\text{即 } x + x^{-1} - xx^{-1} = 0 \Rightarrow x^{-1}(x-1) = 0$$

$$x \neq 1 \text{ 时, } x^{-1} = \frac{x}{x-1} \in I,$$

$$\therefore x = 0 \text{ 时, } x^{-1} = 0$$

$$x = 2 \text{ 时, } x^{-1} = 2.$$

5. ①不可交换, $\forall x, y \in N, x * y = x, y * x = y,$

$$x \neq y \text{ 时, } x * y \neq y * x, \text{ 故不可交换}$$

②可结合. $\forall x, y, z \in N, x * y * z \in N,$

$$x * y * z = x * z = x = x * (y * z) = x * y = x$$

③左单位元: $\forall x \in N, e_1 * x = e_1,$

$$\text{故 } e_1 * x = x \text{ 不恒成立, 无左单位元}$$

④右单位元: $\forall x \in N, x * e_1 = x$

$$\text{故 } \forall x \in N, x \text{ 是右单位元}$$

⑤不存在左单位元, 故无单位元

9. ① $\because *$ 可结合. $\therefore (a * a) * a = a * (a * a)$

$$\text{又 } a * b = b * a \Rightarrow a = b, \therefore a * a = a$$

$$(2) (a * a) * (a * b * a) = a * a * b * a = a * b * a$$

$$(a * b * a) * (a * a) = a * b * (a * a) = a * b * a$$

$$\therefore a * b * a = a * a = a$$

$$(3) (a * c) * (a * b * c) = (a * c * a) * (b * c)$$

$$= a * (b * c) = a * b * c$$

$$(a * b * c) * (a * c) = (a * b) * (c * a * c)$$

$$= (a * b) * c = a * b * c$$

$$\therefore a * b * c = a * c$$